

## SAFETY DATA SHEET

Creation Date 11-October-2010

Revision Date 30-March-2024

Revision Number 4

### 1. Identification

**Product Name** (Chloromethylene)dimethylammonium chloride

**Cat No. :** L12971

**CAS-No** 3724-43-4

**Synonyms** Vilsmeier Reagent

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Corrosive to metals</b>                              | Category 1   |
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Reproductive Toxicity</b>                            | Category 1B  |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |
| <b>Physical Hazards Not Otherwise Classified</b>        | Category 1   |
| Reacts violently with water                             |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

May be corrosive to metals  
Harmful if swallowed  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May damage the unborn child  
Reacts violently with water

**Precautionary Statements****Prevention**

Do not allow contact with water  
Keep container tightly closed  
Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep only in original container  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
Absorb spillage to prevent material damage

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in corrosive resistant polypropylene container with a resistant inliner  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                              | CAS-No    | Weight % |
|--------------------------------------------------------|-----------|----------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | <100     |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

## 5. Fire-fighting measures

|                                         |                                                                                                                                   |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                          |
| <b>Flash Point</b>                      | No information available                                                                                                          |
| <b>Method -</b>                         | No information available                                                                                                          |
| <b>Autoignition Temperature</b>         | Not applicable                                                                                                                    |
| <b>Explosion Limits</b>                 |                                                                                                                                   |
| <b>Upper</b>                            | No data available                                                                                                                 |
| <b>Lower</b>                            | No data available                                                                                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                          |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                        |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Do not expose spill to water.                                                                      |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation. Do not allow contact with water. |
| <b>Storage.</b> | Corrosives area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. To maintain product quality. Keep refrigerated. Incompatible Materials. Bases. Water. Strong oxidizing agents. Amines. Metals. Butyl rubber.                                 |

## 8. Exposure controls / personal protection

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>  | This product does not contain any hazardous materials with occupational exposure limitsestablished by the region specific regulatory bodies.                                                                                                                                                                                                                                                                                                              |
| <b>Engineering Measures</b> | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

### Personal protective equipment

|                        |                   |
|------------------------|-------------------|
| <b>Eye Protection</b>  | Goggles           |
| <b>Hand Protection</b> | Protective gloves |

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                       |              |
|-----------------------|--------------|
| <b>Physical State</b> | Powder Solid |
|-----------------------|--------------|

|                                        |                          |
|----------------------------------------|--------------------------|
| Appearance                             | Yellow                   |
| Odor                                   | pungent                  |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | 139 °C / 282.2 °F        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | Reacts with water        |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | C3 H7 Cl2 N              |
| Molecular Weight                       | 128                      |

## 10. Stability and reactivity

|                                  |                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                   |
| Stability                        | Unstable. Moisture sensitive.                                                                         |
| Conditions to Avoid              | Temperatures above 50°C. Incompatible products. Exposure to moist air or water. Exposure to moisture. |
| Incompatible Materials           | Bases, Water, Strong oxidizing agents, Amines, Metals, Butyl rubber                                   |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                              |
| Hazardous Reactions              | None under normal processing. Reacts violently with water.                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Mist LC50   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                              | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------------|-----------|------------|------------|------------|------------|------------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3131  
 Proper Shipping Name Water-reactive solid, corrosive, n.o.s.  
 Technical Name Methanaminium, N-(chloromethylene)-N-methyl-, chloride  
 Hazard Class 8  
 Packing Group II

### TDG

UN-No UN3261  
 Proper Shipping Name CORROSIVE SOLID, ACIDIC, ORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group II

### IATA

UN-No UN3261  
 Proper Shipping Name CORROSIVE SOLID, ACIDIC, ORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group II

### IMDG/IMO

UN-No UN3261

Proper Shipping Name CORROSIVE SOLID, ACIDIC, ORGANIC, N.O.S.  
 Hazard Class 8  
 Packing Group II

## 15. Regulatory information

### International Inventories

| Component                                              | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS    | NLP |
|--------------------------------------------------------|-----------|-----|------|------|-----------------------------------------------|--------|-----------|-----|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | -   | X    | X    | ACTIVE                                        | -      | 425-970-6 | -   |

| Component                                              | CAS-No    | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------------|-----------|-------|------|------|------|------|------|-------|-------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | -     | -    | -    | -    | X    | -    | X     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component                                              | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | -                                                                   | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                              | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                              | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Methanaminium, N-(chloromethylene)-N-methyl-, chloride | 3724-43-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date**

11-October-2010

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30-March-2024

**Print Date**

30-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**